

1. Two resistance are given as $R_1 = (10 \pm 0.5) \Omega$ and $R_2 = (15 \pm 0.5) \Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is
[2023 (06 Apr Shift 1)]
(1) 6.33
(2) 2.33
(3) 5.33
(4) 4.33
2. The length of a metallic wire is increased by 20% and its area of cross-section is reduced by 4%. The percentage change in resistance of the metallic wire is
[2023 (06 Apr Shift 1)]
3. A cylindrical wire of mass $(0.4 \pm 0.01) \text{ g}$ has length $(8 \pm 0.04) \text{ cm}$ and radius $(6 \pm 0.03) \text{ mm}$. The maximum error in its density will be
[2023 (08 Apr Shift 1)]
(1) 3.5%
(2) 5%
(3) 1%
(4) 4%
4. Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is:
[2023 (08 Apr Shift 1)]
(1) $\frac{\sqrt{5}A}{4}$
(2) $\frac{\sqrt{5}A}{2}$
(3) $\frac{5A}{2}$
(4) $\frac{\sqrt{5}A^2}{2}$
5. A physical quantity P is given as $P = \frac{a^2 b^3}{c \sqrt{d}}$. The percentage error in the measurement of a , b , c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be
[2023 (10 Apr Shift 1)]
(1) 13%
(2) 16%
(3) 12%
(4) 14%
6. When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be:
[2023 (11 Apr Shift 2)]
(1) $\sqrt{5}$
(2) 3
(3) $\sqrt{6}$
(4) $\sqrt{33}$
7. Three forces $F_1 = 10 \text{ N}$, $F_2 = 8 \text{ N}$, $F_3 = 6 \text{ N}$ are acting on a particle of mass 5 kg. The forces F_2 and F_3 are applied perpendicularly so that particle remains at rest. If the force F_1 is removed, then the acceleration of the particle is
[2023 (12 Apr Shift 1)]
(1) 7 m s^{-2}
(2) 0.5 m s^{-2}
(3) 4.8 m s^{-2}
(4) 2 m s^{-2}
8. A vector in $x - y$ plane makes an angle of 30° with y -axis. The magnitude of y -component of vector is $2\sqrt{3}$. The magnitude of x -component of the vector will be :
[2023 (15 Apr Shift 1)]
(1) $\frac{1}{\sqrt{3}}$
(2) 6
(3) 2
(4) $\sqrt{3}$

ANSWER KEYS

1. (4) 2. (25) 3. (4) 4. (2) 5. (1) 6. (4) 7. (4) 8. (3)

1. (4)

For a parallel combination the equivalent resistance is

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \dots (i)$$

A general expression for error can be written as $x \pm \Delta x$

Differentiating equation (i)

$$\frac{dR}{R^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2} \dots (ii)$$

The equivalent resistance with $R_1 = 10 \Omega$ & $R_2 = 15 \Omega$ is

$$\frac{1}{R} = \frac{1}{10} + \frac{1}{15}$$

$$\Rightarrow R = \frac{150}{25} = 6\Omega$$

Substituting the above value in equation (ii)

$$dR = 36 \left(\frac{0.5}{100} + \frac{0.5}{15 \times 15} \right) = 0.26$$

$$\Rightarrow R = 6 \pm 0.26$$

Hence, the percentage error is

$$\frac{dR}{R} \times 100 = \frac{0.26}{6} \times 100 = 4.33\%$$

2. (25)

The given data is

$$l' = 1.2l$$

$$A' = 0.96A$$

Using the formula $R = \frac{\rho l}{A}$, the new resistance is

$$R' = \frac{1.2\rho l}{0.96A}$$

$$\Rightarrow R' = 1.25R$$

$$\Rightarrow \Delta R = 1.25R - R = 0.25R$$

Hence, the percentage increase is

$$= \frac{\Delta R}{R} \times 100$$

$$= \frac{0.25R}{R} \times 100 = 25\%$$

3. (4)

The formula to calculate the density of the material of the wire is given by

$$\rho = \frac{M}{\pi R^2 L} \dots (1)$$

Hence, the formula to calculate the percentage error in calculating the density of the material can be written as

$$\frac{d\rho}{\rho} \times 100 = \left[\frac{dM}{M} + \frac{2dR}{R} + \frac{dL}{L} \right] \times 100 \dots (2)$$

Substitute the values of the known parameters into equation (2) to calculate the required percentage error.

$$= \left[\frac{0.01}{0.4} + \frac{2 \times 0.03}{6} + \frac{0.04}{8} \right] \times 100$$

$$= 4\%$$

4. (2)

Using parallelogram law of vector addition, the formula can be written as

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$$

The data given is

$$|\vec{A}_1| = A, |\vec{A}_2| = \frac{A}{2}$$

Since the vectors are perpendicular $\cos 90^\circ = 0$

$$\Rightarrow F_{\text{net}} = \sqrt{A^2 + \frac{A^2}{4}} = \frac{\sqrt{5}A}{2}$$

5. (1)

The given quantity is $P = \frac{a^2 b^3}{c \sqrt{d}}$.

The percentage error for a quantity can be given by the formula $\frac{\Delta x}{x} \times 100$.

Relation of the relative errors will be,

$$\left(\frac{\Delta P}{P}\right) = \left[\left(\frac{2\Delta a}{a}\right) + \left(\frac{3\Delta b}{b}\right) + \left(\frac{\Delta c}{c}\right) + \left(\frac{1}{2} \frac{\Delta d}{d}\right)\right]$$

Hence, the percentage error is calculated by substituting the values,

$$\left(\frac{\Delta P}{P}\right) \times 100 = (2 \times 1) + (3 \times 2) + 3 + \left(\frac{1}{2} \times 4\right) = 13\%.$$

6. (4)

It is given that $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$.

By the given condition,

$$\vec{B} - (2\hat{i} + 3\hat{j} + 2\hat{k}) = 2\hat{j}$$

$$\Rightarrow \vec{B} = 2\hat{i} + 5\hat{j} + 2\hat{k}$$

The magnitude is given by the formula, $|\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$

$$\Rightarrow |\vec{B}| = \sqrt{4 + 25 + 4} = \sqrt{33}$$

7. (4)

When the force F_1 is removed, the particle starts accelerating because of the resultant force between F_2 and F_3 .

The resultant force can be calculated as follows:

$$F = \sqrt{F_2^2 + F_3^2} \\ = 10 \text{ N}$$

Hence, the acceleration of the particle is given by

$$a = \frac{10 \text{ N}}{5 \text{ kg}} \\ = 2 \text{ m s}^{-2}$$

8. (3)

The angle made by the vector on x axis is

$$\theta = 90^\circ - 30^\circ = 60^\circ$$

Let the vector be $\vec{a}$.

It is given that

$$a_y = 2\sqrt{3}$$

$$\Rightarrow a \sin 60^\circ = 2\sqrt{3}$$

$$\Rightarrow \frac{a\sqrt{3}}{2} = 2\sqrt{3}$$

$$\Rightarrow a = 4$$

x component is

$$a_x = a \cos 60^\circ$$

$$\Rightarrow a_x = 4 \times \frac{1}{2} = 2$$